

FROM RELIABLE TEXT TO REAL VOICES: TRUST-AWARE PROGRESSIVE ADAPTATION FOR LOW-RESOURCE TTS

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ABSTRACT

Adapting multilingual text-to-speech (TTS) models to low-resource languages requires learning target-language pronunciation while retaining zero-shot voice cloning with limited transcribed speech. Synthetic speech and real recordings with automatic speech recognition (ASR) pseudo-labels offer scalable supervision, but differ in transcript reliability and acoustic diversity. We find that their training order affects the balance between content accuracy and speaker similarity, while noisy pseudo-labels can undermine pronunciation learned from synthetic speech. We propose synthetic-to-real adaptation with transcript-agreement weighting: synthetic pairs first establish text–speech correspondences, followed by real recordings to restore speaker conditioning, with agreement between two ASR systems determining each example’s loss weight. Experiments with FireRedTTS3 on Burmese and Lao, and OmniVoice on Burmese, show improved content accuracy with high naturalness and competitive speaker similarity. Jointly considering supervision order and reliability when combining synthetic and real speech offers a practical path to low-resource TTS adaptation with less manual transcription. Audio demos are available at <https://insiderx-pro.github.io/S2R-Adaptation-TTS/>.

Index Terms— Low-resource TTS, synthetic-to-real adaptation, pseudo-label reliability, zero-shot voice cloning

1. INTRODUCTION

Multilingual TTS now clones unseen voices from short reference recordings [1, 2, 3, 4]. CosyVoice 3 broadens acoustic domains [5], and OmniVoice supports over 600 languages, but data imbalance and variable transcript and acoustic quality persist [6]. Low-resource adaptation must therefore learn pronunciation while retaining reference-speaker control from limited supervision [7, 8].

Unpaired text and speech support semi-supervised pretraining [9]; existing models can use synthetic pairs or ASR-labeled recordings. Synthetic pretraining expands text coverage [10, 11] but can reduce speaker similarity, motivating ZeSTA’s domain-conditioned training [12]. CosyVoice 2 scales multilingual training through ASR pseudo-labeling and filtering [13]. Their limitations are complementary: limited teacher voices versus noisy labels.

Combining these sources must preserve pronunciation and voice control. Unlike joint TTS–ASR cycle training with speaker consistency and step-wise optimization [14], we keep ASR fixed and study language adaptation rather than speaker personalization. Synthetic

adaptation improves content accuracy but can reduce speaker similarity; real-speech adaptation restores similarity but noisy labels can erode content gains. Reversing the order changes this balance.

We therefore propose trust-aware progressive adaptation: S-stage adaptation learns text–speech correspondences from synthetic pairs; R-stage adaptation recovers speaker conditioning from real recordings. Agreement between two ASR systems determines loss weights, leaving primary transcripts unchanged. We evaluate FireRedTTS3 [15] on Burmese and Lao and OmniVoice on Burmese, with controlled ablations, independent ASR evaluation, and reference-conditioning and acoustic-history interventions.

Our contributions are:

- **Analysis.** We characterize how supervision order affects content accuracy and speaker similarity.
- **Method.** We propose synthetic-to-real adaptation with transcript-agreement weighting for noisy pseudo-labels.
- **Validation.** We demonstrate effectiveness across languages and backbones, supported by controlled interventions.

2. METHOD

In Fig. 1, the Base model, Synthetic-Stage-Model, and Final-Stage-Model are abbreviated Base, S, and S→R: synthetic supervision precedes reliability-weighted real supervision.

2.1. Synthetic supervision

Fixed Teacher TTS T maps text x_i and voice v_i to $y_i^S = T(x_i, v_i)$. Filtering (Section 3) and same-voice reference pairing yield $\mathcal{D}_S = \{(x_i, y_i^S, r_i^S)\}$, with reference speech r_i^S , example index i , and S/R denoting synthetic/real data. Training uses unit weights and original input text; filtering does not guarantee correct pronunciation.

2.2. Transcript agreement

For real recording y_i^R , fixed primary ASR₁, A_1 , supplies $\tilde{x}_i = A_1(y_i^R)$; fixed secondary ASR₂, A_2 , supplies only a comparison transcript. Training uses $\mathcal{D}_R = \{(\tilde{x}_i, y_i^R, r_i^R)\}$, without transcript fusion or correction. Reference speech r_i^R is a distinct 1–10 s utterance from the same local speaker within a recording.

We measure normalized character disagreement:

$$d_i = \frac{\text{ED}(\mathcal{N}(\tilde{x}_i), \mathcal{N}(A_2(y_i^R)))}{\max(1, |\mathcal{N}(\tilde{x}_i)|)}. \quad (1)$$

ED and $|\cdot|$ denote code-point Levenshtein distance and text length.

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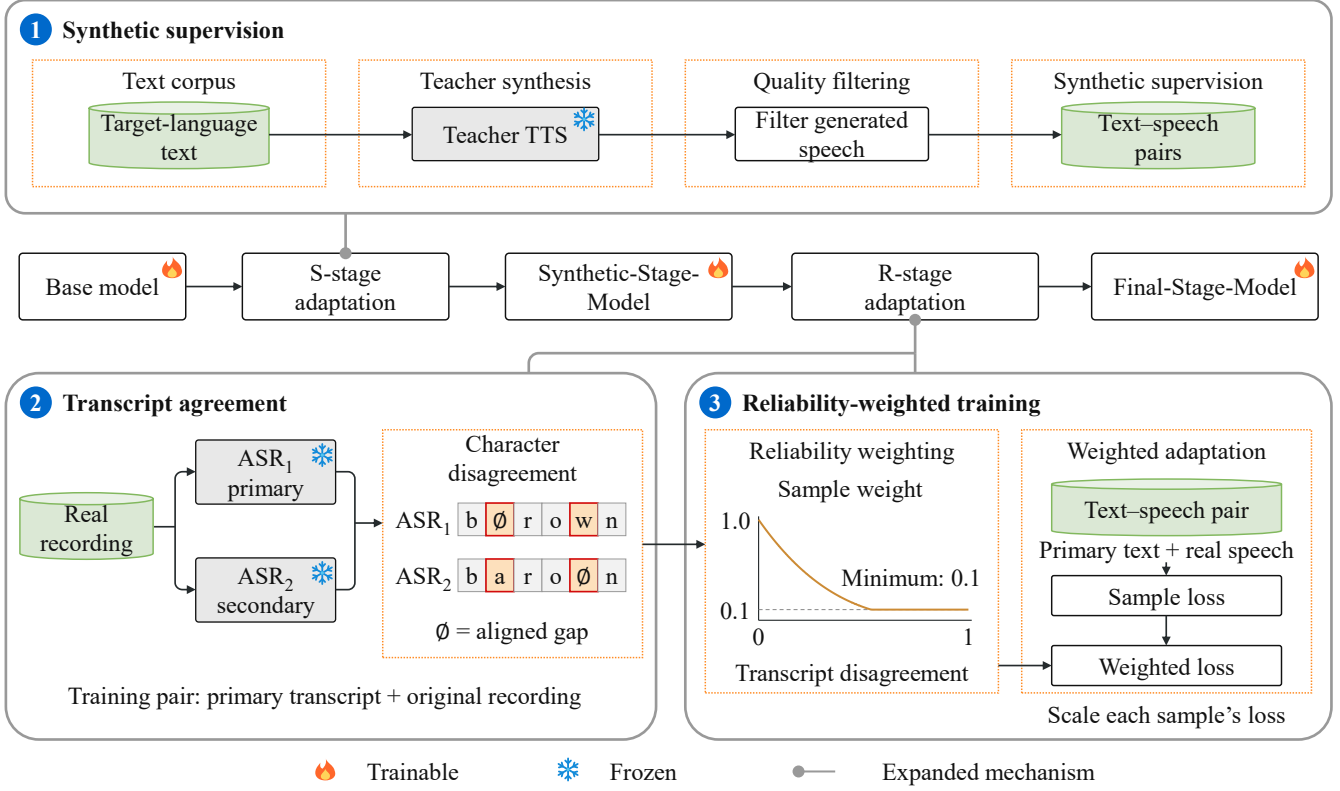


Fig. 1. Trust-aware progressive adaptation. Synthetic supervision establishes text-speech correspondences in S-stage adaptation; transcript agreement guides reliability-weighted training on real recordings in R-stage adaptation. The primary transcript remains the training text.

Scoring-only normalization $\mathcal{N}$ applies Unicode NFC, removing whitespace and punctuation/symbol/control categories but retaining letters, digits, and combining marks, without case, numeral, or Zawgyi conversion. Empty primary labels are excluded; valid empty secondary transcripts give $d_i = 1$, distinct from failed ASR requests.

Agreement does not verify correctness: recognizers may share errors. In Fig. 1, character/transcript disagreement denotes d_i ; $\emptyset$ marks an alignment gap, not a training character.

2.3. Reliability-weighted training

Reliability weighting. To limit noisy supervision’s contribution [16], we assign

$$w_i^{(\gamma)} = \max(w_{\min}, [1 - \min(d_i, 1)]^\gamma), \quad (2)$$

where $\gamma > 0$ controls decay and $w_{\min}$ is the weight floor. Cubic weighting uses $\gamma = 3$, $w_{\min} = 0.10$ (selection: Section 3), giving perfect agreement unit weight while retaining high-disagreement examples with reduced influence.

Weighted adaptation. For minibatch $\mathcal{B}$ with $|\mathcal{B}|$ examples and backbone sample loss $\ell_\theta(y_i^R, \tilde{x}_i, r_i^R)$ at trainable parameters θ ,

$$\mathcal{L}_R(\theta) = \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} w_i^{(\gamma)} \ell_\theta(y_i^R, \tilde{x}_i, r_i^R), \quad (3)$$

averages by example count, not weight sum. Weighting changes loss scale and contributions, but need not scale AdamW updates proportionally. Scoring adds no inference-time ASR. Let $\mathcal{A}_B(\theta; \mathcal{D}, w)$ perform B updates from parameters θ on data $\mathcal{D}$ with weights w .

From Base θ_0 , the stages yield $\theta_S = \mathcal{A}_{B_S}(\theta_0; \mathcal{D}_S, \mathbf{1})$ and $\theta_{SR}^{(\gamma)} = \mathcal{A}_{B_R}(\theta_S; \mathcal{D}_R, w^{(\gamma)})$, where B_S/B_R are update budgets and $1/w^{(\gamma)}$ collect unit/real-example weights. Architectures and objectives stay unchanged; optimizers and schedules restart. R starts from Base; R→S reverses stages.

2.4. Backbone objectives and reference conditioning

FireRedTTS3 [15] updates the generative core, freezing RedAE and CAM++ [17]. Its complete loss $\ell_i = \ell_{\text{flow},i} + 0.1\ell_{\text{stop},i}$ combines flow matching [18] and stop prediction and is weighted as a whole. OmniVoice retains its masked acoustic-token prediction objective [6] with the same weighting.

FireRedTTS3’s frozen RedAE and CAM++ encoders (omitted from Fig. 1) extract acoustic prompt $z^r = E_{\text{AE}}(r)$ and speaker embedding $e^r = E_{\text{spk}}(r)$ from reference speech r , respectively. Controls fix pairings and reference text, exclude ineligible targets, and pair mixture references within each domain. For text x and K patches, its patch-autoregressive distribution is [19, 20]

$$p_\theta(z_{1:K} \mid x, z^r, e^r) = \prod_{k=1}^K p_\theta(z_k \mid x, z^r, e^r, z_{<k}), \quad (4)$$

Model distribution p_θ conditions on history $z_{<k}$: ground-truth patches during training, generated patches at inference. Although the reference encoders are frozen, adaptation can change how the generative core uses their outputs. Reference swaps and matched-prefix interventions (Section 4) probe the generative core’s response to the reference and its dependence on acoustic history, respectively.

Table 1. Main results. (a) CER (%), SIM-O, and H (0–100); adapted CER/SIM-O: mean $\pm$ sample SD over three seeds. (b) MOS (1–5): mean [approximate 95% crossed-bootstrap CI]. (c) Independent-ASR CER (%). Bold: numerical bests, not significance.

(a) Strategy	FireRedTTS3 / Burmese			FireRedTTS3 / Lao			OmniVoice / Burmese		
	CER $\downarrow$	SIM-O $\uparrow$	H $\uparrow$	CER $\downarrow$	SIM-O $\uparrow$	H $\uparrow$	CER $\downarrow$	SIM-O $\uparrow$	H $\uparrow$
Base	142.52	0.7244	0.00	100.45	0.7343	0.00	10.35	0.7278	80.34
S	17.43 $\pm$ 0.45	0.4513 $\pm$ 0.0041	58.36	13.49 $\pm$ 0.26	0.6108 $\pm$ 0.0006	71.61	8.21 $\pm$ 0.20	0.6864 $\pm$ 0.0010	78.54
R	54.09 $\pm$ 0.68	0.7314 $\pm$ 0.0005	56.41	37.06 $\pm$ 0.36	0.7537 $\pm$ 0.0019	68.60	10.74 $\pm$ 0.41	0.7328 $\pm$ 0.0013	80.48
R $\rightarrow$ S	14.71 $\pm$ 0.62	0.3636 $\pm$ 0.0038	50.98	10.64 $\pm$ 0.09	0.5959 $\pm$ 0.0014	71.50	7.36 $\pm$ 0.12	0.6948 $\pm$ 0.0035	79.41
S $\rightarrow$ R: 1.0	23.27 $\pm$ 0.15	0.6942 $\pm$ 0.0014	72.89	14.99 $\pm$ 0.25	0.6993 $\pm$ 0.0010	76.74	10.21 $\pm$ 0.22	0.7142 $\pm$ 0.0017	79.56
S $\rightarrow$ R: 0.5	20.91 $\pm$ 0.37	0.6996 $\pm$ 0.0005	74.24	14.60 $\pm$ 0.14	0.6968 $\pm$ 0.0026	76.74	8.03 $\pm$ 0.33	0.7094 $\pm$ 0.0009	80.10
S$\rightarrow$R: cubic	16.90 $\pm$ 0.26	0.6997 $\pm$ 0.0016	75.97	13.97 $\pm$ 0.36	0.6968 $\pm$ 0.0045	77.00	6.85 $\pm$ 0.08	0.7098 $\pm$ 0.0026	80.57

Strategy	FireRedTTS3 / Burmese		FireRedTTS3 / Lao		OmniVoice / Burmese	
	(b) MOS $\uparrow$	(c) CER $\downarrow$	(b) MOS $\uparrow$	(c) CER $\downarrow$	(b) MOS $\uparrow$	(c) CER $\downarrow$
Base	2.17 [1.97, 2.38]	103.56	2.55 [2.34, 2.76]	100.45	4.40 [4.23, 4.56]	12.62
S	3.15 [2.97, 3.34]	20.45	3.39 [3.22, 3.56]	18.03	4.23 [4.05, 4.40]	11.63
R	1.87 [1.69, 2.06]	68.64	2.12 [1.93, 2.32]	44.94	2.35 [2.15, 2.56]	12.81
R $\rightarrow$ S	2.72 [2.52, 2.92]	17.93	3.27 [3.07, 3.47]	17.13	4.32 [4.12, 4.50]	11.65
S $\rightarrow$ R: 1.0	3.78 [3.55, 4.00]	26.82	3.78 [3.61, 3.94]	23.85	4.35 [4.17, 4.51]	11.69
S $\rightarrow$ R: 0.5	3.91 [3.71, 4.11]	25.06	3.84 [3.65, 4.02]	23.33	4.42 [4.24, 4.58]	11.41
S$\rightarrow$R: cubic	4.12 [3.93, 4.30]	19.34	3.97 [3.77, 4.17]	18.53	4.51 [4.35, 4.66]	9.83

3. EXPERIMENTS

3.1. Experimental setup

Data. Burmese S/R comprise 160,918 Azure Nilar/Thiha pairs (291.148 h) and 48,043 DVB pairs (92.263 h); Lao has 119,246 S and 26,004 R pairs (200.004/45.117 h). ASR₁ is Gemini 2.5 Flash; ASR₂ is Omnilingual ASR [21]. Synthetic filtering checks valid audio/text, 0.4–20 s duration, ASR CER $\leq$ 20%, speech fraction $\geq$ 0.65, clipping $\leq$ 0.005, deduplication, evaluation exclusions, reference eligibility, and a 1,024-token limit. Burmese S starts from 171,240 candidates; 717 real targets lack references. Real speech spans 300 recordings, 1,200 local speaker groups, and 286 unverified cross-recording identity clusters.

Training. Burmese S/R use 53,640/16,015 updates at learning rates $2 \times 10^{-6}/10^{-6}$; Lao uses 39,749/8,668 updates. Orders share AdamW [22], 3% warmup and cosine decay. All adaptation/control runs use one device, accumulation/effective batch 3, and training seeds 42/17/73. Tables 1a and 3 report adapted CER/SIM-O as mean $\pm$ sample SD over these seeds; Base has no training-seed SD. Dev120 (60 texts $\times$ two references) selects $\gamma = 3$, $w_{\min} = 0.10$ by minimum three-seed mean CER with SIM-O $\geq$ 0.6970, over $\gamma \in \{0.5, 1, 2, 3, 4, 6\}$, $w_{\min} \in \{0, 0.05, 0.10, 0.20\}$. Broadcast R uses 16,015 updates, not test-selected.

Evaluation. Burmese Common400 (100 general/300 cloning conditions; 150 texts) uses Omnilingual ASR Unlimited 7B v2. Clone300 uses 30 Burmese/English/Chinese references; SIM-O averages ECAPA-TDNN/WavLM-large cosines [23, 24]. Lao uses Gemini on 404 FLEURS [25] rows (260 inputs, one reference). CER comparisons are within-language. Common400 reuses FLEURS train/development conditions. Normalized evaluation texts do not overlap archived DVB or Lao S/R transcripts. Burmese hashes match none of the 100 checked target/reference conditions. Audited Clone300 speakers are disjoint from adaptation speakers; audio near-duplicate auditing remains incomplete. FireRedTTS3 decoding uses seed 42, 10 flow steps, guidance 2.0, and stop threshold 0.5;

long/empty outputs remain scored. Joint score $H = 200as/(a + s)$ uses $a = \text{clip}(1 - \text{CER}/100, 0, 1)$, $s = \text{clip}(\text{SIM-O}, 0, 1)$, and $H = 0$ if $a + s = 0$.

Listening test. Per language, 20 listeners rated all 30 matched text-reference conditions per system (naturalness 1–5; 600 ratings/system). Burmese backbones share listeners and conditions. Approximate 95% percentile CIs use 10,000 crossed-bootstrap replicates [26], independently resampling listeners and conditions with common indices across systems within each language. We report paired cubic-minus-uniform-0.5 differences. Listening-test audio uses each adapted system’s seed-42 checkpoint.

3.2. Main results

Table 1 shows order dependence, also in matched-batch Lao: uniform S $\rightarrow$ R restores similarity but can worsen CER; R $\rightarrow$ S favors CER. Cubic has the highest observed H and mean MOS. Against uniform 0.5, paired MOS differences are $+0.210$ [0.025, 0.392], $+0.130$ [−0.047, 0.315], and $+0.090$ [−0.052, 0.230] for FireRedTTS3/Burmese, FireRedTTS3/Lao, and OmniVoice/Burmese. Only the first marginal 95% interval excludes zero; these exploratory comparisons are unadjusted for multiplicity. We do not claim consistent MOS improvement across settings. OmniVoice exceeds R in H by just 0.08/100.

Independent ASR. Dolphin-small [27] (Burmese) and XLS-R Lao [28] (Lao) rescore the same generated audio and target texts. Neither participated in pseudo-labeling, reliability estimation, filtering, or model selection. Failures, empty transcripts, and overlong outputs are not selectively excluded. Cubic improves on uniform 1.0 in all groups (Table 1c); R $\rightarrow$ S still yields lower FireRedTTS3 CER, consistent with an accuracy–similarity trade-off.

Adapted OmniVoice leads the shared-text Burmese comparison (Table 2), reducing Base CER by 3.50 points; fixed-voice SIM-O is descriptive. All outputs are scored, including 210/400 mmSpeech Tacotron cases reaching its 12.54 s limit and IMS-Toucan outputs with Burmese phoneme warnings.

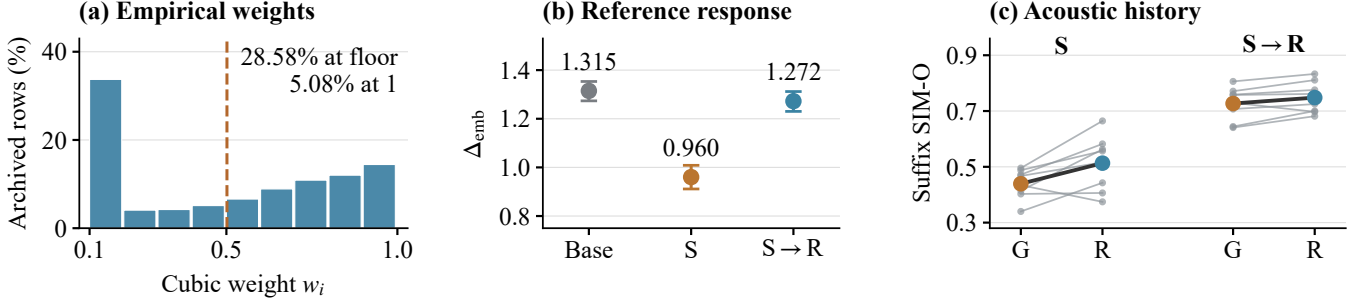


Fig. 2. (a) Cubic weights: 42,785 archived ASR rows. (b) Embedding response: 16 texts, paired-bootstrap 95% intervals. (c) Eight suffix SIM-O pairs: generated (G)/recorded (R) prefixes; thick lines: means. (b,c): historical uniform 0.5 S→R, not cubic.

Table 2. Burmese systems on Common400 / Clone300. *: fixed voices; their SIM-O scores are descriptive.

System	CER (%) ↓	SIM-O ↑
Fish Audio S2-Pro [29]	83.89	0.6376
MMS-TTS-mya* [30]	20.49	0.1029
F5-Myanmar-TTS v2 [31, 32]	35.24	0.5863
IMS-Toucan [33]	95.40	0.3298
mmSpeech Tacotron* [34]	78.70	0.0914
OmniVoice Base [6]	10.35	0.7278
FireRedTTS3 cubic	16.90	0.6997
OmniVoice cubic	6.85	0.7098

Table 3. Burmese FireRedTTS3 controls from S. Mean $\pm$ sample SD over three seeds; shuffles are averaged within seed.

Setting	CER (%) ↓	SIM-O ↑
<i>(a) FLEURS label control; S: Table 1a</i>		
Pseudo-labels, $w = 0.5$	21.17 ± 0.38	0.6345 ± 0.0024
Pseudo-labels, $w = 1.0$	23.67 ± 0.19	0.6487 ± 0.0009
Official text, $w = 1.0$	15.45 ± 0.15	0.6342 ± 0.0012
<i>(b) Broadcast weighting; uniform 1.0: Table 1a</i>		
Uniform 0.5	20.91 ± 0.37	0.6996 ± 0.0005
Linear ($\gamma = 1$)	23.25 ± 0.49	0.6956 ± 0.0030
Global shuffle	21.34 ± 0.52	0.6984 ± 0.0010
Duration-strat. shuffle	21.13 ± 0.22	0.6988 ± 0.0032
Uniform, equal mean	21.09 ± 0.55	0.6993 ± 0.0008
Cubic ($\gamma = 3$)	16.90 ± 0.26	0.6997 ± 0.0016

4. ABLATION AND ANALYSIS

4.1. Supervision order and pseudo-label reliability

Label quality. Table 3a compares labels on identical FLEURS recordings: 3,058 (12.135 h), retaining 2,921 (11.578 h) after overlap/duration exclusions. Official text reduces CER by 8.22 points at $w = 1$, retaining similarity gains; halving pseudo-label weight helps less. Label noise contributes alongside other domain effects.

Reliability validation. On these 2,921 Burmese FLEURS utterances, ASR₁ pseudo-label CER against official text, e_i , correlates with d_i : Spearman $\rho = 0.560$, Pearson $r = 0.546$. Mean label CER is 5.94% at $d_i < 0.1$ ($n = 971$) versus 32.30% at $d_i \geq 0.6$ ($n = 286$). Agreement is informative, not proof of correctness.

Weight assignment. Table 3b fixes initialization, targets, references, order, and updates. Global shuffles of cubic weights (101/202/303 crossed with three training seeds) break assignment; duration-decile shuffles control length correlations. Both preserve weights yet, like equal-mean uniform weighting, underperform cubic: assignment matters beyond loss scale. The 42,785-row archive (Fig. 2a) has mean/SD 0.50166/0.33337 and quartiles 0.10000/0.54771/0.81090. Its 588 later-excluded rows and missing later transcripts distinguish it from the 48,043-target pool (mean 0.500946). The pool’s duration-weighted mean 0.58715 becomes 0.50110/ $\approx$ 0.5865 with global/duration-stratified shuffling.

Mixtures. A 0/10/25/50/75/90/100% real-example sweep from S uses 16,015 updates with unit S/cubic R weights; synthetic replay restores less similarity than real-only continuation. Joint training from Base matches 160,920 S/48,045 R exposures and the learning-rate schedule (restart at 53,640), but yields 20.160% CER/0.65120 SIM-O, supporting progressive adaptation under matched exposure.

4.2. Reference conditioning and acoustic history

Reference response. For 16 texts, we independently swap acoustic prompts and speaker embeddings between two same-text references. With $q = \text{SIM}(g, r_A) - \text{SIM}(g, r_B)$ for generated speech g (SIM-O cosines), Δ_{emb} averages the r_B -to- r_A embedding-induced change in q over prompts/texts. Base/S/historical uniform 0.5 S→R score 1.3147/0.9602/1.2723 (Fig. 2b); matched-reference SIM-O scores are 0.7804/0.5672/0.7625. Conflicts follow embeddings; R restores S-weakened response; cubic is untested.

History dependence. Eight of 23 prefix pairs share normalized ASR text at CER $\leq 5\%$. Fixing reference, boundary, and subsequent random state, recorded rather than generated prefixes improve suffix SIM-O by 0.0739/0.0211 for S/historical uniform 0.5 S→R (Fig. 2c), in 7/8 cases each. Paired 95% intervals [0.0225, 0.1204]/[0.0025, 0.0374] condition on cases, not seeds. Scoring excludes prefixes and the first 250 ms; unchanged-history replay reproduces outputs. This shows history dependence, not naturally accumulating voice drift.

5. CONCLUSION

We proposed transcript-agreement weighting for synthetic-to-real TTS adaptation. Across FireRedTTS3 (Burmese/Lao) and OmniVoice (Burmese), supervision order and reliability shape the accuracy–similarity trade-off. Label controls identify transcription noise; shuffled and equal-mean controls support informative sample assignment beyond loss scale. Reference and history interventions link the trade-off to conditioning response, with evidence specific to the evaluated settings. Organizing weak supervision by order and reliability supports low-resource voice cloning with high naturalness, competitive speaker similarity, and less manual transcription.

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